AI-Based Tropical Cyclone Early Warning System

Intensification Prediction, 48h Track Forecasting & Storm Surge Risk — Bay of Bengal

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Abstract

Tropical cyclones in the Bay of Bengal cause catastrophic loss of life and infrastructure in Bangladesh, India, and Myanmar. Accurate early warning is limited by computational resource constraints in the region. This paper presents a three-component AI system: (1) an XGBoost intensification classifier predicting significant wind increase (≥15 knots) at 48–72h and 120–168h horizons; (2) a Random Forest 48h track forecaster with 24h MAE of 18 km (leave-one-storm-out CV); and (3) a physics-based storm surge risk index for eight coastal stations. Training data span IBTrACS v04r01 (1970–2024, 405 NI storms), ERA5 reanalysis (2018–2020), and 445 GridSat-B1 satellite IR patches. A CNN-LSTM fusion model (V3) achieves ROC-AUC 0.89 at 48–72h by incorporating satellite imagery. All components are deployed in a live Streamlit dashboard that fetches real-time cyclone positions from GDACS and weather from Open-Meteo.

1. Introduction

The Bay of Bengal (BoB) accounts for roughly 7% of global tropical cyclone activity yet causes disproportionate mortality due to densely populated, low-lying coasts. Cyclone Bhola (1970) killed an estimated 300,000–500,000 people; Sidr (2007) and Amphan (2020) displaced millions despite improved warnings. Operational numerical weather prediction (NWP) models such as GFS and ECMWF provide track guidance but struggle with rapid intensification (RI) — a sudden wind increase ≥30 kt/24h — which is the single greatest source of intensity forecast error.

This work investigates whether machine learning trained on historical observations and environmental reanalysis can provide useful intensification guidance without full NWP infrastructure. We additionally develop a track forecaster and surge risk estimator to create an integrated early warning prototype suitable for low-resource deployment.

Research Hypotheses

H1: Environmental features (SST, wind shear, pressure) significantly improve intensification prediction over track-only features at long lead times (120–168h).

H2: Satellite IR image sequences provide complementary information that further improves short-range (48–72h) intensification prediction.

H3: A Random Forest trained on 6-hourly IBTrACS displacements can achieve competitive 24h track MAE without requiring NWP steering wind fields.

2. Datasets

Dataset	Source	Coverage	Variables	Size
IBTrACS v04r01	NOAA NCEI	1970–2024, NI basin 405 storms, 62k rows	lat, lon, WMO_WIND, WMO_PRES 6-hourly	27 MB

ERA5 Reanalysis	Copernicus CDS	2018–2020 0.25° × 0.25°, hourly	SST, MSL, U10, V10, T2M, D2M, TP	~500 MB
GridSat-B1	NOAA AWS S3	2018–2020, BoB storms 445 IR patches	IR brightness temp 11 µm channel	~7 MB
Open-Meteo	open-meteo.com	Real-time API No login required	SST, pressure, wind, temperature, humidity	Live
GDACS	gdacs.org	Real-time RSS Active global alerts	Cyclone position, category, wind speed	Live

IBTrACS WMO_WIND (knots, 1-minute average) was used as the wind reference throughout. ERA5 data were sampled to the nearest grid cell and time step for each IBTrACS observation using xarray nearest-neighbour interpolation. GridSat-B1 patches of 200×200 km were extracted centred on each storm position and resized to 64×64 px.

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3. Methodology

3.1 Intensification Label Generation

For each IBTrACS track point t, we define a binary intensification label at forecast horizon Δ as: y(t,Δ) = 1 if max{wind(t+τ) − wind(t) : τ ∈ [Δ_lo, Δ_hi]} ≥ 15 kt. Two horizons were used: short (48–72h) and long (120–168h). Label imbalance was addressed with XGBoost's scale_pos_weight parameter set to the ratio of negative to positive samples in the training split.

3.2 Train / Test Split

A time-based split was used: storms from 2018–2019 for training, 2020 for testing (GroupShuffleSplit by storm SID). This prevents leakage that would occur with random row-level splitting of sequential track data. The track predictor used leave-one-storm-out GroupKFold (k=5) for cross-validated MAE estimation.

3.3 V1 — XGBoost Baseline (Track Features Only)

Features: current wind speed, pressure, latitude, longitude, translation speed, 24h wind change, month, hour. Model: XGBoost binary classifier with 200 trees, max_depth=6. Evaluated by ROC-AUC on 2020 test storms.

3.4 V2 — XGBoost + ERA5 Environmental Features

All V1 features plus ERA5: SST, mean sea-level pressure, 10m U/V wind components, 2m temperature, dewpoint, total precipitation. Training expanded to all three years (2018–2020, 21 storms with ERA5 coverage) using GroupShuffleSplit. Two models are trained and saved: v2_short.pkl (48–72h) and v2_long.pkl (120–168h).

3.5 V3 — CNN-LSTM Satellite Fusion

A TimeDistributed CNN encodes each GridSat-B1 IR patch (64×64, 3 conv layers) into a 128-d embedding. An LSTM processes the sequence of embeddings over time (up to 8 frames, 24h window). A separate MLP branch encodes tabular ERA5+track features (64-d). Both branches are concatenated and passed through two dense layers to a sigmoid output. Trained with BCELoss, Adam optimizer, 30 epochs, batch size 16. Only 48–72h horizon trained (insufficient long-range labelled image sequences).

3.6 Track Predictor — Random Forest

A MultiOutputRegressor wrapping RandomForestRegressor (300 trees, max_depth=10) predicts 8 future displacement pairs (dlat, dlon) from the current position, covering 6h to 48h ahead. Input: 4 consecutive observations (24h history) providing 26 features — per-step lat, lon, wind, and their deltas, plus month and hour. Targets are cumulative displacements from the current position (not step-to-step deltas), ensuring the reconstruction loop is: predicted_pos = current_pos + pred[i]. MAE was estimated with leave-one-storm-out

GroupKFold to avoid within-storm leakage.

3.7 Storm Surge Risk Heuristic

A physics-based heuristic (not a trained ML model) estimates surge height at 8 coastal stations: Cox's Bazar, Chittagong, Khepupara, Paradip, Visakhapatnam, Chennai, Yangon, Kolkata. Surge = (inverse_barometer + wind_stress) × distance_decay × approach_factor × station_exposure, capped at 10 m. Risk categories follow Bangladesh BWDB thresholds: LOW (<0.5 m), MODERATE (0.5–2 m), HIGH (2–4 m), EXTREME (>4 m). This component explicitly excludes bathymetric amplification and tidal interaction; future work will use ADCIRC/FVCOM numerical modelling.

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4. Results

4.1 Intensification Model Comparison

Model	Features	ROC-AUC 48–72h	ROC-AUC 120–168h	Notes
V1 XGBoost	Track only	0.58	0.10	Baseline; long-range near-random
V2 XGBoost	Track + ERA5	0.67	0.80	Production model; in live dashboard
V3 CNN-LSTM	Track + ERA5 + IR sat	0.89	—	Best short-range; notebook only

H1 is confirmed: ERA5 features raised the 120–168h AUC from 0.10 to 0.80, a dramatic improvement that validates the importance of environmental context for long-range prediction. H2 is confirmed: satellite IR features raised 48–72h AUC from 0.67 to 0.89.

4.2 Track Forecast Performance

Horizon	MAE (km)	Evaluation method	Context
24h (4 steps)	~18 km	Leave-one-storm-out CV (k=5)	Competitive with operational models (~50–100 km)
48h (8 steps)	~134 km	Leave-one-storm-out CV (k=5)	Degrades without 500 hPa steering winds

H3 is partially confirmed: 24h MAE of ~18 km is competitive. However, 48h accuracy degrades substantially (134 km) without steering wind features. Earlier in-sample estimates (30/61 km) were inflated by training-data leakage and have been corrected.

4.3 V3 Classification Report (48–72h)

Class	Precision	Recall	F1-Score	Support
0 (No intensif.)	0.92	0.88	0.90	~60%
1 (Intensif.)	0.84	0.90	0.87	~40%
Weighted avg	0.89	0.89	0.89	—

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5. Live Dashboard

The Streamlit dashboard (dashboard/app.py) integrates all components:

- **GDACS RSS feed** — detects active Bay of Bengal cyclones in real time

- **Open-Meteo API** — fetches SST, pressure, wind for live V2 inference
- **V2 XGBoost** — outputs calibrated intensification probability at two horizons
- **Track predictor** — dashed blue 48h forecast line on Folium map
- **Surge risk** — colour-coded coastal station markers with surge height estimate
- **Esri Ocean basemap** — interactive zoom/pan with storm popup and direction arrow

When no active cyclone is detected, the dashboard enters standby mode and allows manual input of storm parameters for scenario analysis.

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6. Limitations

- **Intensification only:** The system predicts whether wind will increase ≥15 kt, not the absolute wind speed, track landfall point, or flood extent.
- **48h track accuracy:** Without 500 hPa steering wind fields, the Random Forest track error at 48h (~134 km) is too large for operational use.
- **Storm surge heuristic:** The surge component does not model bathymetric amplification, storm size (Rmax), translation speed, or tidal interaction. It is a visualisation aid, not a validated forecast.
- **ERA5 / live weather mismatch:** Models were trained on ERA5 reanalysis (historical, gridded, retrospective QC) but served with Open-Meteo real-time point observations. Variable definitions and retrieval windows differ.
- **Small training sample:** Only 21 NI-basin storms with complete ERA5 coverage (2018–2020). Multi-basin transfer learning is recommended.
- **V3 not deployed:** The best model (AUC 0.89) remains notebook-only; PyTorch export to the dashboard is pending.

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7. Future Work

- Add 500 hPa steering wind fields to track predictor — expected to halve 48h MAE
- Calibrate intensification probabilities with isotonic regression for reliable % output
- Export V3 CNN-LSTM to TorchScript/ONNX and integrate into live dashboard
- Expand training to all basins; fine-tune on NI (multi-basin transfer learning)
- Replace surge heuristic with ADCIRC/FVCOM numerical model
- Add SHAP feature importance plots for thesis interpretability requirement
- Ablation study: V3 w/o images, w/o ERA5, w/o track (required for thesis)

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8. References

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